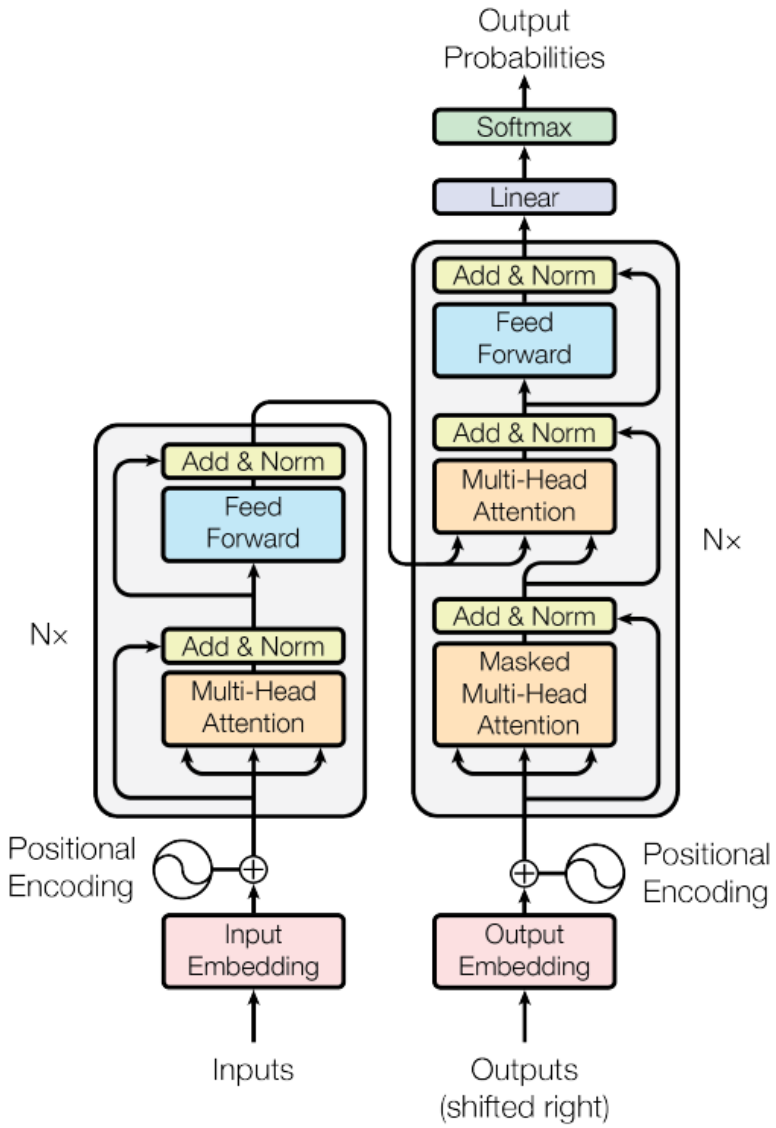
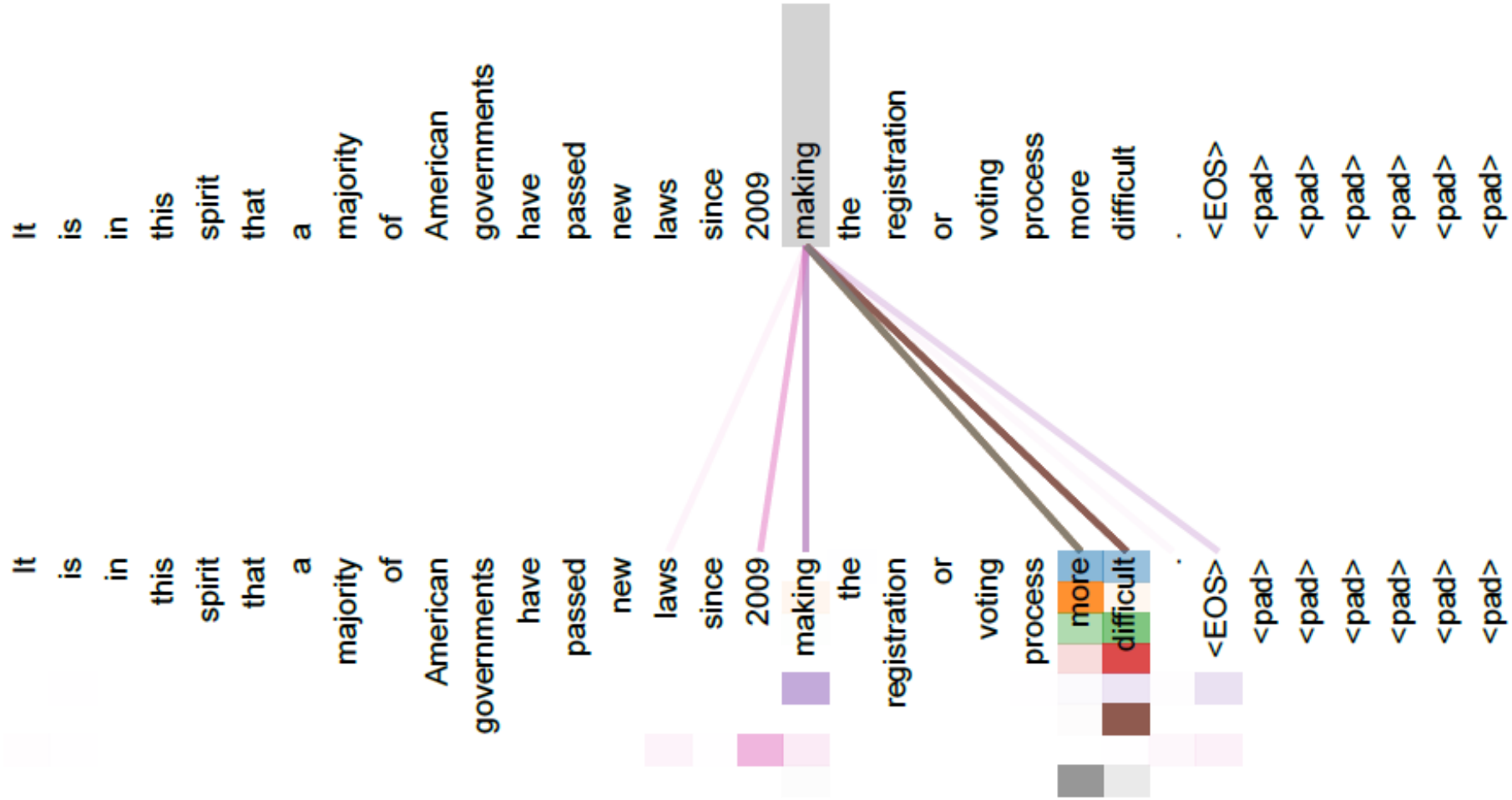
Transformer models

Deep learning

Transformer for machine translation

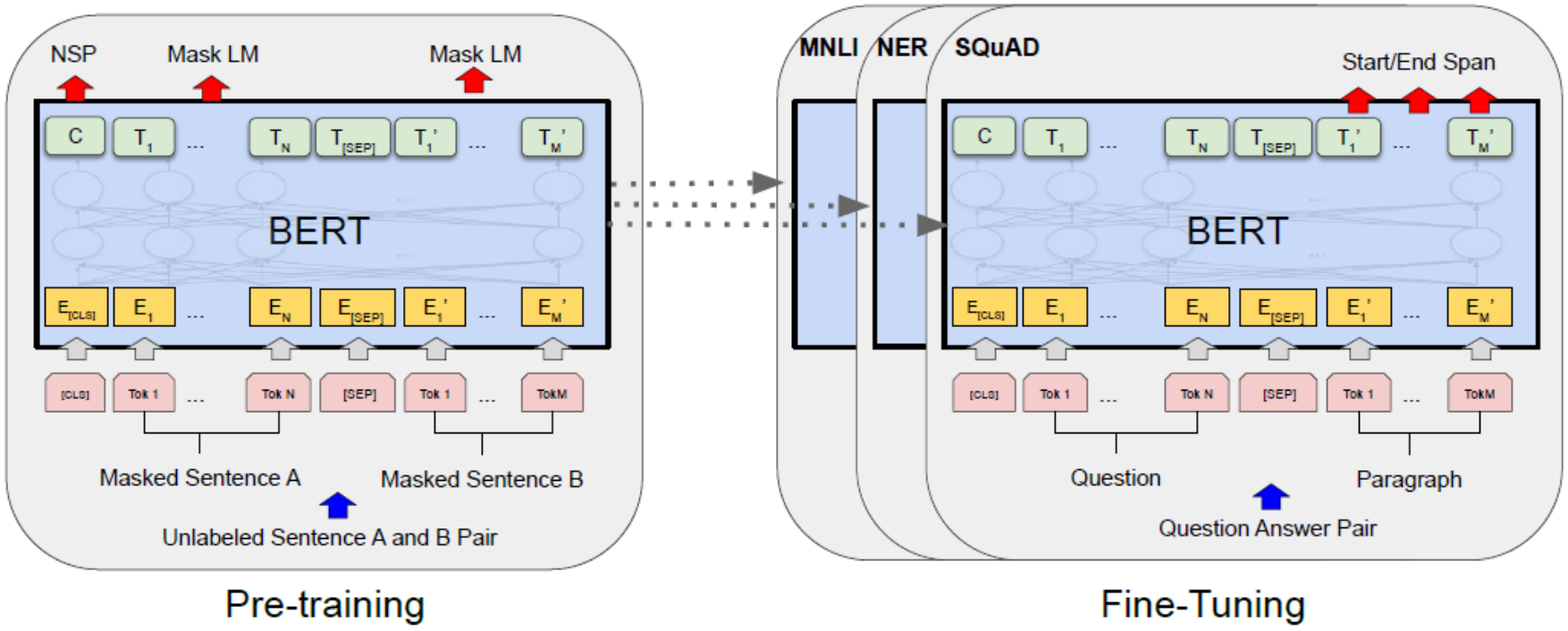


Attention visualization



BERT (Bidirectional Encoder Representation from Transformers, 2018)

1. Using only Transformer encoder
2. Masked Language Model pretraining
3. Next Sentence Prediction for training CLS embedding
4. Fine-tuning for downstream tasks



RoBERTa (2019)

1. Next Sentence Prediction is not used, only Masked Language Model
2. Randomly sampled masks
3. Much larger training dataset

Model	data	bsz	steps	SQuAD (v1.1/2.0)	MNLI-m	SST-2
RoBERTa						
with BOOKS + WIKI	16GB	8K	100K	93.6/87.3	89.0	95.3
+ additional data (§3.2)	160GB	8K	100K	94.0/87.7	89.3	95.6
+ pretrain longer	160GB	8K	300K	94.4/88.7	90.0	96.1
+ pretrain even longer	160GB	8K	500K	**94.6/89.4**	**90.2**	**96.4**
$\text{BERT}_{\text{LARGE}}$						
with BOOKS + WIKI	13GB	256	1M	90.9/81.8	86.6	93.7
$\text{XLNet}_{\text{LARGE}}$						
with BOOKS + WIKI	13GB	256	1M	94.0/87.8	88.4	94.4
+ additional data	126GB	2K	500K	94.5/88.8	89.8	95.6

ModernBERT (2024)

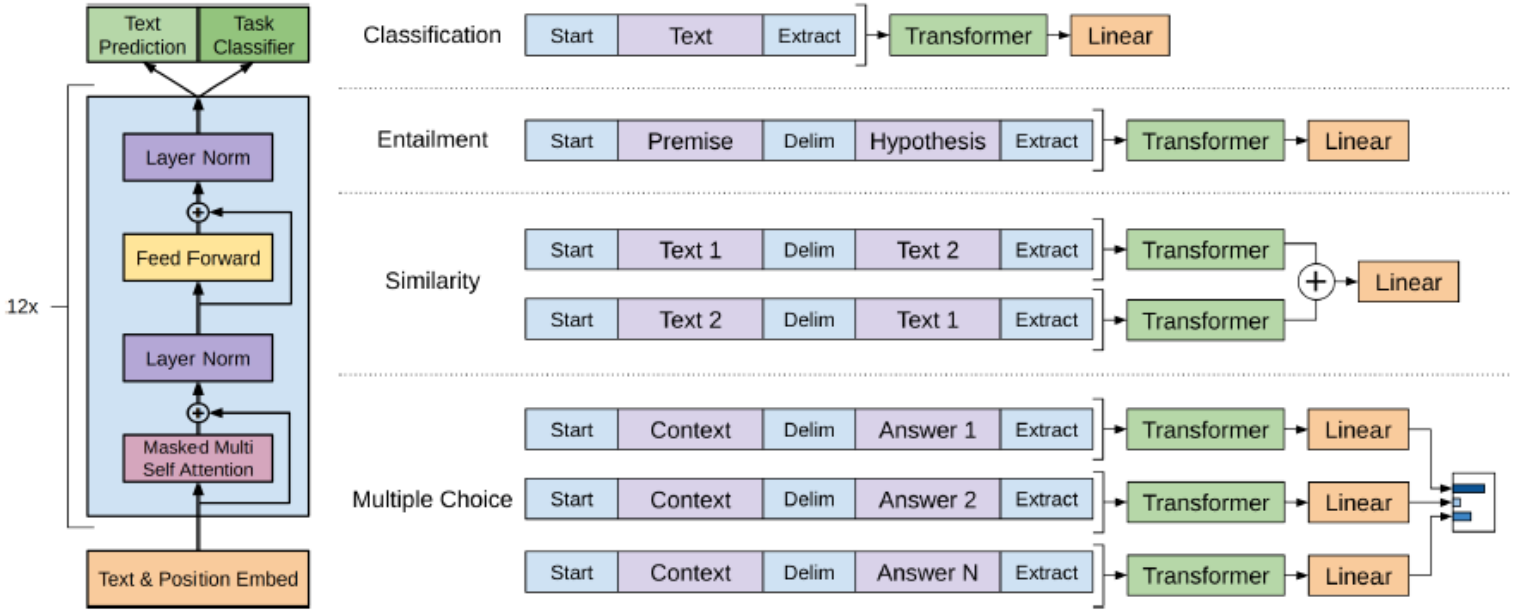
1. Modern transformer architecture: RMSNorm, Gated FFN, RoPE
2. Training: Masked Language Model
3. Local-Global interleaving self-attention: two layers with local attention (window size 128) and one layer with global attention
4. RoPE: for local attention theta=10000, for global attention theta=160000
5. Two-staged training: 1) contexts of length 1024, then 2) contexts of length 8192

GPT-1 (Generative Pretraining from Transformers)

1. Using only Transformer decoder (without cross-attention parts)
2. Unsupervised pretraining using next token prediction
3. Fine-tuning for supervised tasks

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \ldots, u_{i-1}; \Theta)$$

Observation: fine-tuning requires only few epochs



GPT-like vs. BERT-like models

- BERT-like models are better for token embeddings, since they look at both left and right context
- GPT-like models are better for prompt answering, since they are trained in next-token-prediction fashion

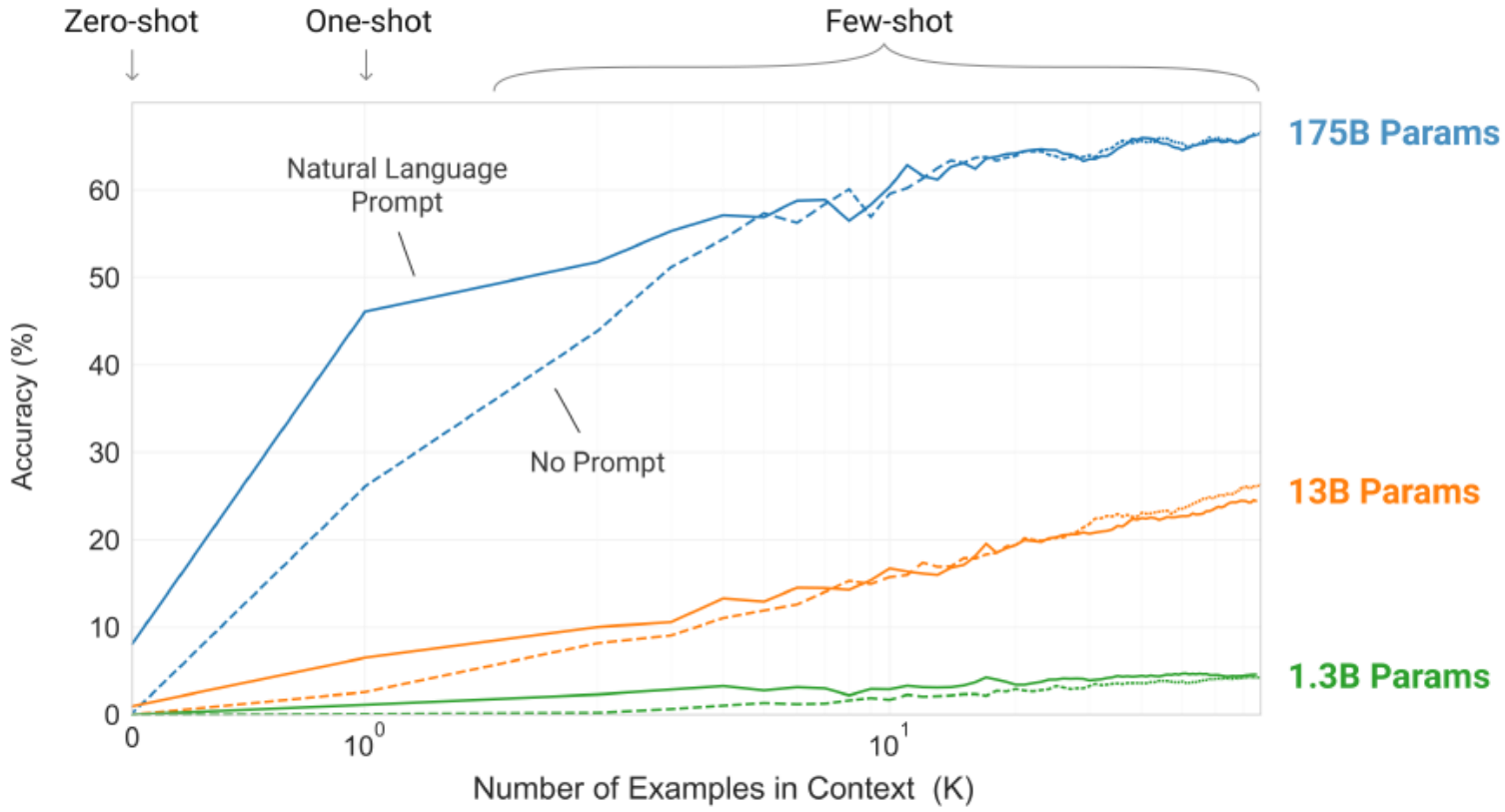
GPT-2

1. Increased sizes of model and dataset
2. Fully unsupervised training
3. Shows not that bad zero-shot performance

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	**21.8**
117M	**35.13**	45.99	**87.65**	**83.4**	**29.41**	65.85	1.16	1.17	37.50	75.20
345M	**15.60**	55.48	**92.35**	**87.1**	**22.76**	47.33	1.01	**1.06**	26.37	55.72
762M	**10.87**	**60.12**	**93.45**	**88.0**	**19.93**	**40.31**	**0.97**	**1.02**	22.05	44.575
1542M	**8.63**	**63.24**	**93.30**	**89.05**	**18.34**	**35.76**	**0.93**	**0.98**	**17.48**	42.16

GPT-3

1. Significantly increased sizes of model and dataset
2. Fully unsupervised training
3. Shows much better zero-shot performance



LLM training pipeline

Stage 1. Pretraining using Next-token prediction

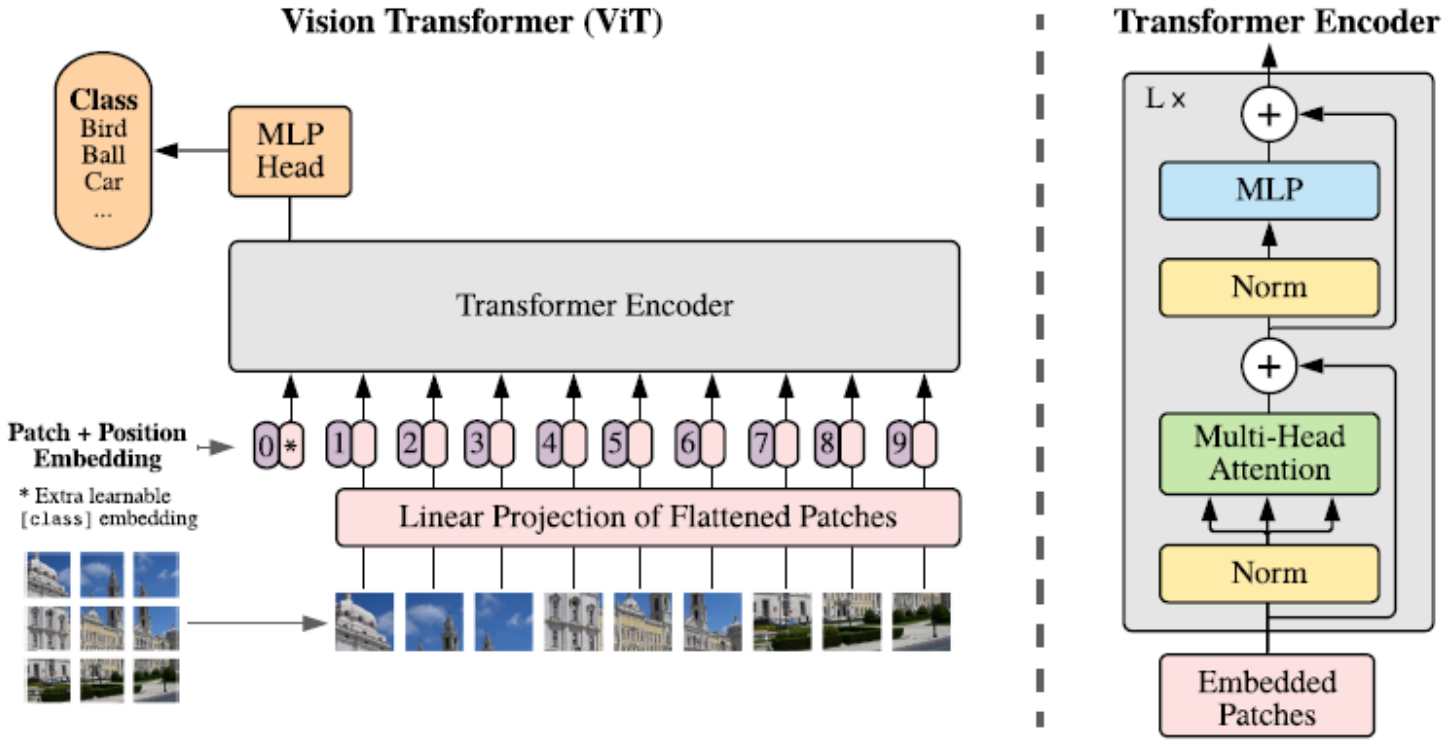
Stage 2. Learning instruction following by SFT – Supervised Fine-Tuning on golden dataset with prompts and good answers

Stage 3. Learning polite answering using DPO (Direct Policy Optimization) with preference data

Stage 4. Learning complex problem solving using RLVR (Reinforcement Learning with Verifiable Rewards)

Vision Transformer

Using similar unsupervised masked pretraining mechanism for images



BERT/GPT models for images

- Vision Transformer – an analogue of BERT/RoBERTa models, allows finding good image representation and is used as alternative to convolutional encoders
- ImageGPT – an analogue of InstructGPT/ChatGPT models of image pixels, can be used for image generation, but in practice diffusion-based models like DALL-E, Stable Diffusion are usually used